Lab Assignment 04

BRAC UNIVERSITY

Inspiring Excellence

Course Code:	**CSE111**
Course Title:	**Programming Language II**
Topic:	**Constructor, Constructor Overloading and Multiclass Problem**
Number of Tasks:	**11 (Coding: 08, Tracing: 03)**

[Submit all the Coding Tasks (Task 1 to 8) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Task 9 to 11) handwritten to your Lab Instructors at the beginning of the lab]

[You are not allowed to change the driver codes of any of the tasks]

Task 1

Design the **Student** class in such a way that it produces the following output.

Driver Code	Expected Output

`public class StudentTester{`
`  public static void main(String[] args){`
`  Student s1 = new Student("Harry", "CSE");`
`  System.out.println(s1.name);`
`  s1.updateName("Harry Potter");`
`  System.out.println(s1.name);`
`  System.out.println(s1.prog);`
`  s1.updateProgram("CS");`
`  String prog = s1.accessProgram();`
`  System.out.println(prog);`
`  }`
`}`
	`Harry`
	`Harry Potter`
	`CSE`
	`CS`

Task 2

Design the **Toy** class in such a way that it produces the following output

Driver Code	Expected Output

`public class ToyTester{`
`  public static void main(String[] args){`
`  Toy t1 = new Toy("Car", 230);`
`  System.out.println("1==========");`
`  t1.updatePrice(250);`
`  System.out.println("2==========");`
`  System.out.println(t1.name);`
`  t1.showPrice();`
`  System.out.println("3==========");`
`  Toy t2 = new Toy("Robot", 450);`
`  System.out.println("4==========");`
`  t2.updateName("Autobot");`
`  System.out.println("5==========");`
`  System.out.println(t2.name);`
`  t2.showPrice();`
`  }`
`}`
	`A new toy has been made!`
	`1==========`
	`2==========`
	`Car`
	`price: 250 Taka`
	`3==========`
	`A new toy has been made!`
	`4==========`
	`Changing old name: Robot`
	`new name: Autobot`
	`5==========`
	`Autobot`
	`price: 450 Taka`

Task 3

Design the **Shape2D** class in such a way that it produces the following output.

Driver Code	Expected Output

`public class Shape2DTester {`
`  public static void main(String[] args) {`
`  Shape2D sq = new Shape2D();`
`  System.out.println("---------1----------");`
`  sq.area();`
`  System.out.println("---------2----------");`
`  Shape2D rectangle = new Shape2D(5,6);`
`  System.out.println("---------3----------");`
`  rectangle.area();`
`  System.out.println("---------4----------");`
`  Shape2D tri1 = new Shape2D(5,6,"Triangle");`
`  System.out.println("---------5----------");`
`  tri1.area();`
`  System.out.println("---------6----------");`
`  Shape2D tri2 = new Shape2D(5,6,7);`
`  System.out.println("---------7----------");`
`  tri2.area();`
`  System.out.println("---------8----------");`
`  }`
`}`
	`A Square has been created with`
	`length: 5`
	`---------1----------`
	`The area of the Square is: 25.0`
	`---------2----------`
	`A Rectangle has been created with`
	`length: 5 and breadth:  6`
	`---------3----------`
	`The area of the Rectangle is: 30.0`
	`---------4----------`
	`A Triangle has been created with`
	`height: 5 and base:  6`
	`---------5----------`
	`The area of the Triangle is: 15.0`
	`---------6----------`
	`A Triangle has been created with`
	`the following sides: 5, 6, 7`
	`---------7----------`
	`The area of the Triangle is: 14.69`
	`---------8----------`

Task 4

Write "**Student**" class to show the following expected outputs

Note:

- A student can't take any course until the CGPA is set.
- A student cannot take more than 4 courses.
- A student with CGPA below 3 cannot take more than 3 courses.

Driver Code	Expected Output

`public class StudentDriver {`
`  public static void main(String[] args){`
`  Student student1 = new Student(12345678);`
`  System.out.println("1----------");`
`  student1.addCourse("CSE110");`
`  System.out.println("2----------");`
`  student1.storeCG(2.5);`
`  student1.addCourse("CSE110");`
`  student1.addCourse("ENG101");`
`  student1.showAdvisee();`
`  System.out.println("3----------");`
`  student1.removeAllCourse();`
`  student1.showAdvisee();`
`  System.out.println("4----------");`
`  student1.storeID(54652365);`
`  String[] courses = {"SOC101","CSE111","ENG102"};`
`  student1.addCourse(courses);`
`  student1.showAdvisee();`
`  System.out.println("5----------");`
`  student1.addCourse("CSE230");`
`  student1.showAdvisee();`
`  System.out.println("6----------");`
`  Student student2 = new Student(975738383,3.7);`
`  System.out.println("7----------");`
`  String[] courses2 =`
`{"CSE220","PHY112","MAT120","BUS101","CHN101"};`
`  student2.addCourse(courses2);`
`  student2.showAdvisee();`
`  }`
`}`
	`A student with ID 12345678 has`
	`been created.`
	`1----------`
	`Failed to add CSE110`
	`Set CG first`
	`2----------`
	`Student ID: 12345678, CGPA: 2.5`
	`Added courses are:`
	`CSE110 ENG101`
	`3----------`
	`Student ID: 12345678, CGPA: 2.5`
	`No courses added.`
	`4----------`
	`Student ID: 54652365, CGPA: 2.5`
	`Added courses are:`
	`SOC101 CSE111 ENG102`
	`5----------`
	`Failed to add CSE230`
	`CG is low. Can't add more than 3`
	`courses.`
	`Student ID: 54652365, CGPA: 2.5`
	`Added courses are:`
	`SOC101 CSE111 ENG102`
	`6----------`
	`A student with ID 975738383 and`
	`cgpa 3.7 has been created.`
	`7----------`
	`Failed to add CHN101`
	`Maximum 4 courses allowed.`
	`Student ID: 975738383, CGPA: 3.7`
	`Added courses are:`
	`CSE220 PHY112 MAT120 BUS101`

Task 5

Design the **Triangle** Class that will produce the following output. We will consider both triangles to have the same sides if all sides are equal in the same orientation/sequence only.

Types of Triangle:

- Equilateral: When all sides in the same orientation are equal.
- Isosceles: When any two sides of a triangle in the same orientation are equal.
- Scalene: When all sides are of different lengths.

Driver Code	Output

public class TriangleTester{
  public static void main(String args[]){
  Triangle t1 = new Triangle();
  Triangle t2 = new Triangle();
  Triangle t3 = new Triangle();
  Triangle t4 = new Triangle();
 	
  t1.updateSides(4, 4, 4);
  t2.updateSides(4, 5, 6);
  t3.updateSides(4, 5, 6);
  t4.updateSides(5, 4, 6);
 	
  t1.triangleDetails();
  System.*out*.println("--------1--------");
  System.*out*.println(t1.printTriangleType());
  System.*out*.println("--------2--------");
  t3.triangleDetails();
  System.*out*.println(t3.printTriangleType());
  System.*out*.println("--------3--------");
  t4.triangleDetails();
  System.*out*.println(t4.printTriangleType());
  System.*out*.println("--------4--------");
  t2.compareTrinagles(t3);
  System.*out*.println("--------5--------");
  t1.compareTrinagles(t2);
  System.*out*.println("--------6--------");
  t1 = t2;
  t1.compareTrinagles(t2);
  System.*out*.println("--------7--------");
  t3.compareTrinagles(t4);
  }
}
	Three sides of the triangle are: 4, 4, 4
	Perimeter: 12
	--------1--------
	This is an Equilateral Triangle.
	--------2--------
	Three sides of the triangle are: 4, 5, 6
	Perimeter: 15
	This is a Scalene Triangle.
	--------3--------
	Three sides of the triangle are: 5, 4, 6
	Perimeter: 15
	This is a Scalene Triangle.
	--------4--------
	Addresses are different but the sides of the triangles are equal.
	--------5--------
	Addresses, length of the sides and perimeter all are different.
	--------6--------
	These two triangle objects have the same address.
	--------7--------
	Only the perimeter of both triangles is equal.

Task 6

Write the **Teacher** and **Course** classes so that the TestTeacher class produces the outputs given.
Hint: A teacher can add a maximum of 3 courses.

Driver Code	Output

`public class TestTeacher{`
`  public static void main(String [] args){`
`  Teacher t1 = new Teacher("Matin Saad Abdullah","MSA");`
`  Teacher t2 = new Teacher("Mumit Khan","MMK");`
`  Teacher t3 = new Teacher("Sadia Hamid Kazi","SKZ");`
`  Course c1 = new Course("CSE 110");`
`  Course c2 = new Course("CSE 111");`
`  Course c3 = new Course("CSE 220");`
`  Course c4 = new Course("CSE 221");`
`  Course c5 = new Course("CSE 230");`
`  Course c6 = new Course("CSE 310");`
`  Course c7 = new Course("CSE 320");`
`  Course c8 = new Course("CSE 340");`
`  t1.addCourse(c1);`
`  t1.addCourse(c2);`
`  t2.addCourse(c3);`
`  t2.addCourse(c4);`
`  t2.addCourse(c5);`
`  t3.addCourse(c6);`
`  t3.addCourse(c7);`
`  t3.addCourse(c8);`
`  System.out.println("1=======================");`
`  t1.printDetail(); // this.courses[index].code`
`  System.out.println("2=======================");`
`  t2.printDetail();`
`  System.out.println("3=======================");`
`  t3.printDetail();`
`  }`
`}`
	`A new teacher has been created`
	`A new teacher has been created`
	`A new teacher has been created`
	`1=======================`
	`Name: Matin Saad Abdullah`
	`Initial: MSA`
	`List of courses:`
	`CSE 110`
	`CSE 111`
	`2=======================`
	`Name: Mumit Khan`
	`Initial: MMK`
	`List of courses:`
	`CSE 220`
	`CSE 221`
	`CSE 230`
	`3=======================`
	`Name: Sadia Hamid Kazi`
	`Initial: SKZ`
	`List of courses:`
	`CSE 310`
	`CSE 320`
	`CSE 340`

Task 7

Design the required class/es so that the following output is generated. Read the following description:

1. You may assume that to board a bus, a student must have the bus pass, and his/her destination must match the route of the bus.

2. Additionally, the default maximum capacity of the bus is 2.

Driver Code

```
public class BracuStudentTester {
 public static void main(String[] args) {
  BracuStudent st1 = new BracuStudent("Afif", "Mirpur");
  System.out.println("1===============");
  BracuStudent st2 = new BracuStudent("Shanto", "Motijheel");
  BracuStudent st3 = new BracuStudent("Taskin", "Mirpur");
  st1.showDetails();
  st2.showDetails();
  System.out.println("2===============");
  st3.showDetails();
  System.out.println("3===============");
  BracuBus bus1 = new BracuBus("Mirpur");
  BracuBus bus2 = new BracuBus("Azimpur", 5);
  bus1.showDetails();
  bus2.showDetails();
  System.out.println("4===============");
  st2.getPass();
  st3.getPass();
  System.out.println("5===============");
  st2.showDetails();
  st3.showDetails();
  System.out.println("6===============");
  bus1.board();
  System.out.println("7===============");
  bus1.board(st1, st2);
  System.out.println("8===============");
  st1.getPass();
  st2.updateHome("Mirpur");
  st1.showDetails();
  st2.showDetails();
  System.out.println("9===============");
  bus1.board(st1);
  bus1.board(st2, st3);
  System.out.println("10===============");
  bus1.showDetails();
 }
}
```

Output

```
1===============
Student Name: Afif
Lives in Mirpur
Have Bus Pass? false
Student Name: Shanto
Lives in Motijheel
Have Bus Pass? false
2===============
Student Name: Taskin
Lives in Mirpur
Have Bus Pass? false
3===============
Bus Route: Mirpur
Passenger Count: 0 (Max: 2)
Passengers on Board:
Bus Route: Azimpur
Passenger Count: 0 (Max: 5)
Passengers on Board:
4===============
5===============
Student Name: Shanto
Lives in Motijheel
Have Bus Pass? true
Student Name: Taskin
Lives in Mirpur
Have Bus Pass? true
6===============
No passengers
7===============
You don't have a bus pass!
You got on the wrong bus!
8===============
Student Name: Afif
Lives in Mirpur
Have Bus Pass? true
Student Name: Shanto
Lives in Mirpur
Have Bus Pass? true
9===============
Afif boarded the bus.
Shanto boarded the bus.
Bus is full!
10===============
Bus Route: Mirpur
Passenger Count: 2 (Max: 2)
Passengers on Board:
Afif Shanto
```

Task 8

Design the **Student** and the **Usis** class so that the following output is produced.
Note:

- A student's email, password, and login status are null by default while creating an object of the Student class.
- Your code should satisfy the conditions mentioned in the output only.
- Usis class will have two instance variables: totalAdvisee and an array of Student type to store the student object. The array will be updated inside the advising() method only when the advising is successful.
- Usis can take at most 5 advisees.

Driver Code	Expected Output

`public class UsisTester {`
`  public static void main(String[] args) {`
`  Student rakib = new Student("Rakib", 12301455, "CSE");`
`  Student roy = new Student("Roy", 12501345, "CS");`
`  System.out.println("1*************");`
`  Usis usisObj = new Usis();`
`  System.out.println("2*************");`
`  usisObj.login(rakib);`
`  System.out.println("3*************");`
`  usisObj.advising(rakib);`
`  System.out.println("4*************");`
`  rakib.email = "rakib@hotmail.com";`
`  rakib.password = "1234";`
`  System.out.println("5*************");`
`  usisObj.login(rakib);`
`  System.out.println("6*************");`
`  usisObj.advising(rakib);`
`  System.out.println("7*************");`
`  usisObj.advising(rakib, "CSE110", "PHY111", "MAT110", "CSE260");`
`  System.out.println("8*************");`
`  usisObj.advising(rakib, "CSE110", "PHY111","MAT110");`
`  System.out.println("9*************");`
`  usisObj.allAdviseeInfo();`
`  System.out.println("10*************");`
`  roy.email = "roy@hotmail.com";`
`  roy.password = "abcd";`
`  usisObj.login(roy);`
`  System.out.println("11*************");`
`  usisObj.advising(roy, "CSE110", "ENG101", "PHY112");`
`  System.out.println("12*************");`
`  usisObj.allAdviseeInfo();`
`  }`
`}`
	`Student object is created`
	`Student object is created`
	`1*************`
	`Usis is ready to use!`
	`2*************`
	`Email and password need to be set.`
	`3*************`
	`Please login to advise courses!`
	`4*************`
	`5*************`
	`Login successful`
	`6*************`
	`You haven't selected any courses.`
	`7*************`
	`You need special approval to take more than 3 courses.`
	`8*************`
	`Advising successful!`
	`9*************`
	`Total Advisee: 1`
	`Name: Rakib ID: 12301455`
	`Department: CSE`
	`Advised Courses:`
	`CSE110 PHY111 MAT110`
	`==========`
	`10*************`
	`Login successful`
	`11*************`
	`Advising successful!`
	`12*************`
	`Total Advisee: 2`
	`Name: Rakib ID: 12301455`
	`Department: CSE`
	`Advised Courses:`
	`CSE110 PHY111 MAT110`
	`==========`
	`Name: Roy ID: 12501345`
	`Department: CS`
	`Advised Courses:`

Task 9

```
public class B{
  public int x = 3, y = 5, temp = -5, sum = 2;
  public B(){
    y = temp + 3 ;
    sum = 3 + temp + 2;
    temp -= 2;
  }
  public B(B b){
    sum = b.sum;
    x = b.x + 2;
    b.methodB(2,3);
  }
  public void methodA(int m, int n){
    int x = 2;
    y = y + m + (temp++);
    x = x + 5 +  n;
    sum = sum + x + y;
    System.out.println(x + " " + y+ " " + sum);
  }
  public void methodB(int m, int n){
    int  y = 0;
    y = y + this.y;
    x = this.y + 2 + temp;
    methodA(x, y);
    sum = x + y + sum;
    System.out.println(x + " " + y+ " " + sum);
  }
}
```

```
public class Tester9 {
   public static void main(String args []){
    B b1 = new B();
    B b2 = new B(b1);
    b1.methodA(1, 2);
    b2.methodB(3, 2);
   }
}
```

Outputs		

Task 10

```
public class msgClass{
  public int content;
}
class FinalT5A{
  public int sum = 2, y = 1, x = 1;
  public void methodA(){
    int x=6, y =0;
    msgClass myMsg = new msgClass();
    myMsg.content = this.x;
    x = x + myMsg.content;
    this.y = this.y + methodB(myMsg, myMsg.content);
    System.out.println(x + " " + this.y+ " " + sum);
    y =  this.y/2 + this.x;
    x = y + sum/2;
    sum = x + y + myMsg.content;
    System.out.println(x + " " + y+ " " + sum);
  }
  public int methodB(msgClass mg2, int mg1){
    int x = 0;
    y = y + mg2.content;
    mg2.content = y + mg1;
    x = this.x + 3 + mg1;
    sum = sum + x + y;
```

```
    System.out.println(this.x + " " + this.y+ " " + sum);
    mg2.content = sum - mg1 ;
    return sum;
  }
}
```

DRIVER CODE	OUTPUTS		

`public class Tester10{`
`  public static void main(String args []){`
`  FinalT5A fT5A = new FinalT5A();`
`  fT5A.methodA();`
`  }`
`}`			

Task 11

```
public class A{
  public int temp = 3, sum = 9, y = 4, x = 0;
  public A(){
    int sum = 7;
    y = temp - 5;
    sum = temp + 2;
    temp-=2;
    this.x = sum + temp + y;
  }
  public A(int y, int temp){
    y = temp - 1 + x;
    sum = temp + 2 -x;
    temp-=2;
  }
  public void methodA(int m, int [] n){
    int x = 0;
    y = y + m + methodB(x,m);
    x = this.x + 2 + (++n[0]);
    sum = sum + x + y;
    n[0] = sum + 2;
```

```
21      System.out.println(n[0] + " " + y+ " " + sum);
22    }
23    public int methodB(int m, int n){
24      int [] y = {0};
25      this.y = y[0] + this.y + m;
26      x = this.y + 2 + temp - n;
27      sum = x + y[0] + this.sum;
28      System.out.println(y[0]+ " "+ temp + " " + sum);
29      return y[0];
30    }
31  }
```

Driver Code	Output		

public class Tester11 {
public static void main(String args[]){
int[] x = {35};
A a1 = new A();
A a2 = new A(-5,-7);
a1.methodA(1, x);
a2.methodA(1, x);
}
}			

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Design the **Parcel** class in such a way that it produces the following output.

NOTE: For the method ***calcFee()***, if the delivery location is ***Dhanmondi***, then the location charge will be 50 taka or else it'll be free. Also, while calculating total fee, if the product weight is 0 the total_fee would also be 0.

Formula: fee = (weight * 20) + *location_charge (if any)*

Driver Code	Expected Output

`public class ParcelDriver {`
`  public static void main(String[] args){`
`  Parcel p1 = new Parcel();`
`  p1.printDetails();`
`  p1.name = "Spongebob";`
`  p1.printDetails();`
`  System.out.println("1**********");`
`  Parcel p2 = new Parcel("Bob the Builder");`
`  p2.weight = 15;`
`  p2.calcFee("Gulshan");`
`  p2.printDetails();`
`  System.out.println("2**********");`
`  p2.addWeight(25);`
`  p2.calcFee("Banani");`
`  p2.printDetails();`
`  System.out.println("3**********");`
`  Parcel p3 = new Parcel("Dora the Explorer", 10);`
`  p3.addWeight(15);`
`  p3.calcFee("Dhanmondi");`
`  p3.printDetails();`
`  }`
`}`
	`Set name first`
	`Name: Spongebob`
	`Total Weight: 0`
	`Total Fee: 0.0`
	`1**********`
	`Name: Bob the Builder`
	`Total Weight: 15`
	`Total Fee: 300.0`
	`2**********`
	`Updated Weight: 40`
	`Name: Bob the Builder`
	`Total Weight: 40`
	`Total Fee: 800.0`
	`3**********`
	`Updated Weight: 25`
	`Name: Dora the Explorer`
	`Total Weight: 25`
	`Total Fee: 550.0`

Task 2

Design the program to get the output as shown.

Hints:

- Create an array in the Team class to store the player's object
- Use constructor overloading technique for Team class

```java
public class TeamTester {
  public static void main(String[] args) {
    Team b = new Team();
    b.updateName("Bangladesh");
    Player mashrafi =  new Player("Mashrafi", 42, 100);
    b.addPlayer(mashrafi);
    Player tamim =  new Player("Tamim", 35, 70);
    b.addPlayer(tamim);
    b.printDetail();
    System.out.println("==============");
    Team a = new Team("Australia");
    Player ponting =  new Player("Ponting", 50, 300);
    a.addPlayer(ponting);
    Player lee =  new Player("Lee", 49, 200);
    a.addPlayer(lee);
    a.printDetail();
  }
}
```

Output:

```
Team: Bangladesh
List of players:
Name: Mashrafi
Age: 42, Total Matches: 100
Name: Tamim
Age: 35, Total Matches: 70
==============
Team: Australia
List of players:
Name: Ponting
Age: 50, Total Matches: 300
Name: Lee
Age: 49, Total Matches: 200
```

Task 3

```java
public class TracingX {
  public int x, y = 1;
  public int metA(int y){
    y += x + 3;
    int temp = y + this.y;
    if (temp % 2 == 0){
      return temp;
    }
    TracingX t = new TracingX();
    t.y = this.x - (++x) + t.x;
```

```
    this.y = y + t.metA(t.x);
    System.out.println(x +" "+ y +" "+temp);
    return temp+this.y;
  }
}
```

Driver code:

```
public class TesterX {
 public static void main(String[] args) {
   TracingX t1 = new TracingX();
   t1.y = t1.x = 5;
   TracingX t2 = new TracingX();
   t2.x = t1.metA(2);
   t2.y = t2.metA(4);
   System.out.println(t1.y +t1.x +" "+t2.x +" "+t2.y);
 }
}
```

Output: